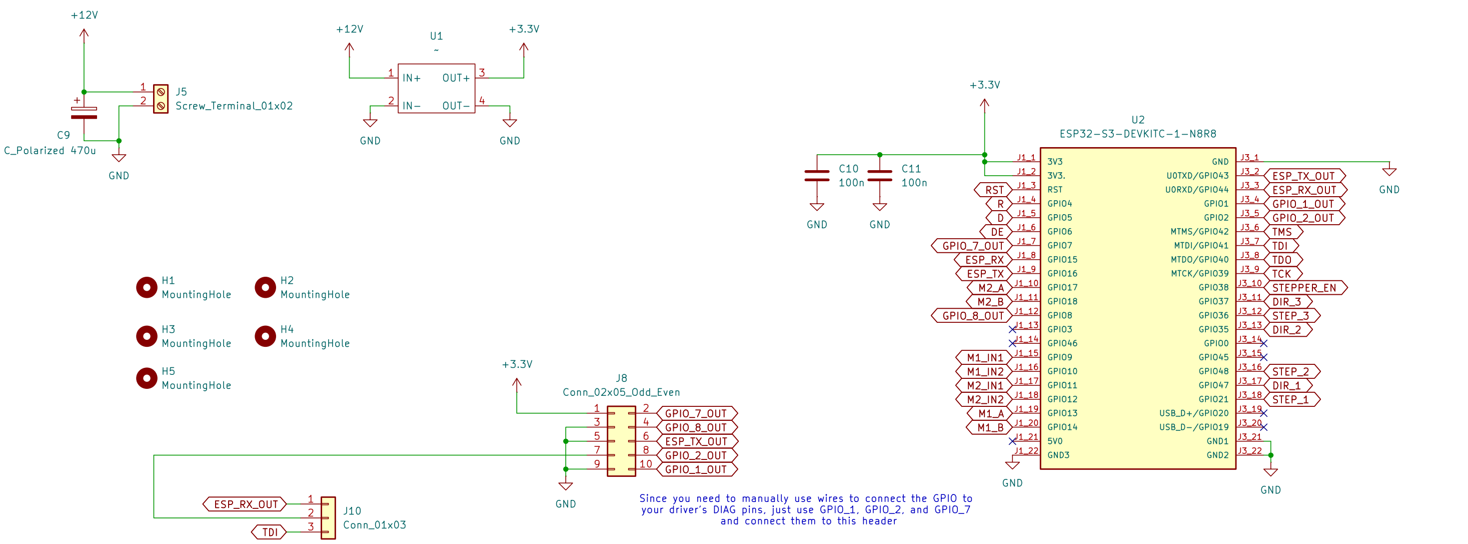


Since you need to manually use wires to connect the GPIO to your driver's DIAG pins, just use GPIO_1, GPIO_2, and GPIO_7 and connect them to this header

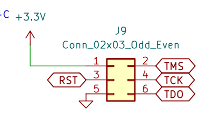
ShamanLink Connections:
1. UART, connect TX–RX and vice versa (matched headers)on the 02x05 header; power ShamanLink, optionally through TagConnect pin 1, but NOT through 02x05 header pin 1. Otherwise through USB–C or by applying 3V3 directly to TagConnect from supply. switch this jumper to ESP_RX_OUT; adjust ShamanLink firmware to utalize PIO0_30 for UART TX.

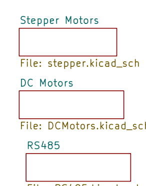
2. JTAG, ShamanLink was designed for SWD but can support JTAG by altering the firmware to do so. The pins used are PIO0_30 for TDI (switch jumper on MotorTotem); and the SWD pins for the rest. Therefore you must use both TagConnect and at least one pin of the 02x05 header.

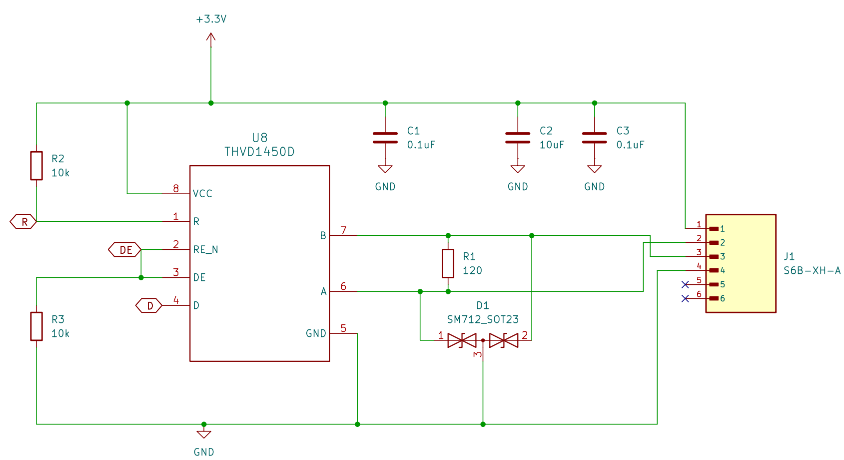
3. Resets, on the TagConnect, the reset pin is pulled high and can be toggled low by the ShamanLink to reset the ESP32. No additional jumpers are required to get this to work. Just power the ShamanLink and connect the TagConnect.

4. RS–485, these connections are straightforward and one to one as far as the connectors are concerned. You will simply have to implement the communication protocol in firmware so the tranceivers work. Optional connection for additional communication over long distances.

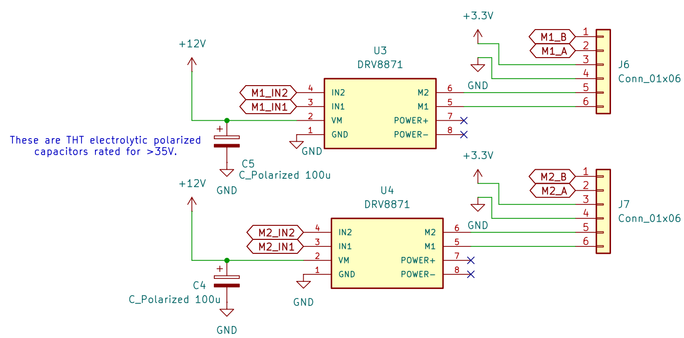
PS. 02x05 header useless as far as communicating to the ShamanLink beyond UART as pins 2, 4, 8, and 10 are connected to the ShamanLink SWD input, if connected to ESP GPIO, it will cause issues. If the 02x05 is completely disconnect there is no consequence. If the RS–485 is completely disconnected, there is no consequence. If the TagConnect is completely disconnected, RST is now floating, TCK is now floating, and TMS is now floating. We prefer for all of the strapping pins to be floating since this is a dev kit and the pull up/down hardware is already there. Our pull ups/downs on TCK and TMS don't cause any issues, but the RST pull up on the ShamanLink creates an equivalent pull up of 5k. This is too strong and will render USB programming incapable. It should however work for JTAG programming. Bottom line: if you aren't using JTAG to program the ESP32, there is no reason to have the TagConnect connected. It only serves that purpose (and powering the ShamanLink).



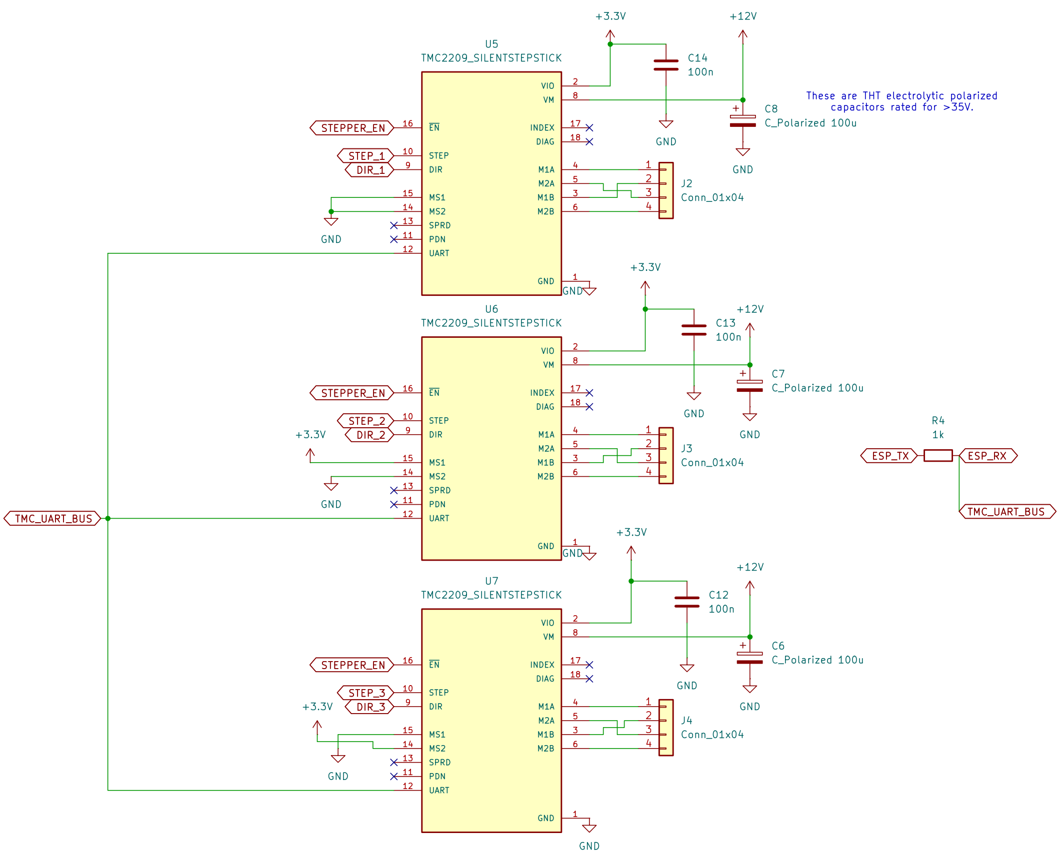




+3.3V
U8
THVD1450D
R2
10k
C1
0.1uF
C2
10uF
C3
0.1uF
GND
GND
GND
8
VCC
1
R
2
RE_N
3
DE
4
D
R
DE
D
B
7
A
6
GND
5
R1
120
D1
SM712_SOT23
1
2
3
R3
10k
GND
J1
S6B-XH-A
1
2
3
4
5
6



These are THT electrolytic polarized capacitors rated for >35V.
+12V
U3
DRV8871
M1_IN2
M1_IN1
IN2
IN1
VM
GND
M2
M1
POWER+
POWER−
C5
C_Polarized 100u
GND
+3.3V
M1_B
M1_A
J6
Conn_01x06
U4
DRV8871
M2_IN2
M2_IN1
C4
C_Polarized 100u
M2_B
M2_A
J7
Conn_01x06


U5
TMC2209_SILENTSTEPSTICK
U6
TMC2209_SILENTSTEPSTICK
U7
TMC2209_SILENTSTEPSTICK
VIO
VM
EN
INDEX
DIAG
STEP
DIR
M1A
M2A
M1B
M2B
MS1
MS2
SPRD
PDN
UART
GND
+3.3V
+12V
C14
100n
C13
100n
C12
100n
C8
C_Polarized 100u
C7
C_Polarized 100u
C6
C_Polarized 100u
These are THT electrolytic polarized
capacitors rated for >35V.
STEPPER_EN
STEP_1
DIR_1
STEP_2
DIR_2
STEP_3
DIR_3
J2
Conn_01x04
J3
Conn_01x04
J4
Conn_01x04
TMC_UART_BUS
R4
1k
ESP_TX
ESP_RX
TMC_UART_BUS